

Charles "Chaz" Merritt

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Education

University of Georgia, BA in Cognitive Science	2020 – 2024
- GPA: 3.5/4.0 Coursework: Search & Optimization, Learning Algorithms, Epistemology	
University of Georgia, MS in Artificial Intelligence	2024 – 2026
Coursework: Reinforcement Learning, Data Science, Advanced Forest Planning Research: Quantifying Economic Value-at-Risk in Timber Simulation Models With Monte Carlo Simulation	

Experience

Researcher, University of Georgia, PERSEUS – Athens, GA	May 2026 – Present
- Developed a landscape scale timber modeling and optimization framework based on variational inference. - Co-authored 3 peer-reviewed papers on AI and forestry. - Developed an ArcGIS toolbox for semi-automated image segmentation and feature extraction.	
Research Assistant, University of Georgia – Athens, GA	Aug. 2024 – May 2026
- Contributed to peer-reviewed papers at the intersection of AI and forestry. - Built a full-stack web interface for optimization workflows (Python, FastAPI, TypeScript) - Applied Monte Carlo simulation and risk modeling (Value-at-Risk, CVaR) to quantify economic uncertainty	
Geospatial Analyst Intern, Northwest Management – Moscow, ID	May – Sep. 2024
- Designed automated GIS pipelines using Python, Rasterio, and GeoPandas. - Implemented clustering algorithms and image segmentation models (SAM) for forest analysis. - Processed and analyzed large-scale spatial datasets for operational forestry decision-making.	

Projects

Optimization Software Web Interface	2026
- Developed an interactive dashboard for the SETS timber supply models. - Tools Used: Railway, FastAPI, Python, TypeScript, HiGHS-Py	
Brandon AI	2026
- Developed contrastive representation models for phonetic similarity learning with InfoNCE loss. - Tools Used: PyTorch, NumPy	

Publications

Thesis: Stochastic Valuation of Planted Loblolly Pine Stands Using Monte Carlo Simulation	May. 2026
Chaz Merritt, Chair: Pete Bettinger	
Peer-Reviewed Article: AI Technology Use in Forestry, Journal of Forestry	Feb. 2026
Krista Merry, Pete Bettinger, Chaz Merritt, Khaled Rasheed, Bruno da Silva, Roger C. Lowe III	

Skills

Programming: Python, TypeScript
Machine Learning: PyTorch, Reinforcement Learning, Contrastive Learning, Optimization